interstellar space. The frequency will be lowered by precisely the same amount as the flow of time is slowed. A lower frequency means a longer wavelength, so light from the star must be shifted toward the red end of the spectrum by the same amount as time is dilated on the star's surface.

At the Sun's surface the time dilation is 2 parts in a million, so the gravitational redshift of light arriving at the Earth from the Sun should also be 2 parts in a million. This was too small a redshift to be measured definitively in Einstein's day, but in the early 1960s, technology began to catch up with Einstein's laws of gravity: Jim Brault of Princeton University, in a very delicate experiment, measured the red-shift of the Sun's light, and obtained a result in nice agreement with Einstein's prediction.

Within a few years after Schwarzschild's untimely death, his spacetime geometry became a standard working tool for physicists and astrophysicists. Many people, including Einstein, studied it and computed its implications. All agreed and took seriously the conclusion that, if the star were rather large in circumference, like the Sun, then spacetime inside and around it should be very slightly curved, and light emitted from its surface and received at Earth should be shifted in color, ever so slightly, toward the red. All also agreed that the more compact the star, the greater must be the warpage of its spacetime and the larger the gravitational redshift of light from its surface. However, few were willing to take seriously the extreme predictions that the Schwarzschild geometry gave for highly compact stars (Figure 3.4):

The Schwarzschild geometry predicted that for each star there is a critical circumference, which depends on the star's mass—the same critical circumference as had been discovered by John Michell and Pierre Simon Laplace more than a century earlier: 18.5 kilometers times the mass of the star in units of the mass of the Sun. If the star's actual circumference is larger than this critical one by a factor of 4 (upper part of Figure 3.4), then the star's space will be moderately curved as shown, time at its surface will flow 15 percent more slowly than far away, and light emitted from its surface will be shifted toward the red end of the spectrum by 15 percent. lf the star's circumference is smaller, just twice the critical one (middle part of Figure 3.4), its space will be more strongly curved, time at its surface will flow 41 percent more slowly than far away, and light from its surface will